Module 07: Communication — Social Cognition, Language, and Intersubjectivity

> **Course**: Active Inference 401 | **Unit**: Philosophical Foundations | **Audience**: Advanced undergraduates / graduate students

Learning Objectives

1. Analyze Active Inference's account of **intersubjectivity** — how minds meet — and compare it with phenomenological, analytic, and developmental accounts.
2. Evaluate the **language as active inference** hypothesis: communication as mutual model alignment.
3. Examine the philosophical implications of **shared generative models** for social ontology, institutions, and collective intentionality.

Introduction

Communication is more than information transfer. Philosophically, it raises fundamental questions: How can one mind understand another? What is language's role in thought? How do social institutions emerge from individual cognition? Active Inference provides a formal framework that connects these questions through the concept of **coupled inference** — agents whose actions serve as observations for each other, driving mutual model alignment.

Key Concepts

1. The Problem of Other Minds

The traditional philosophical problem: How do I know that other people have minds? The evidence is behavioral — I observe their actions — but behavior underdetermines mental states. Classical solutions include:

Argument from analogy (Mill): Others behave like me, and I have a mind, so they probably do too.

Simulation theory: I understand others by simulating their mental states using my own cognitive machinery.
Theory theory: I understand others by applying a folk-psychological theory that maps behavior to mental states.

Active Inference offers a fourth option — **generative model-based inference**:

- I model other agents as systems with their own generative models
- I infer their beliefs, goals, and predictions by observing their actions and treating those actions as evidence
- This is **active inference about active inference** — a higher-order generative model that models another agent's modeling process
- Crucially, this doesn't require "getting inside their head" — it requires having a model good enough to predict their behavior

This connects to developmental psychology: infants develop increasingly sophisticated models of others' mental states (theory of mind) through progressive Bayesian updating, beginning with simple predictions and advancing to full mentalization.

2. Language as Active Inference

The **language as active inference** hypothesis (Friston et al., 2020; Vasil et al., 2020) proposes that linguistic communication is a special case of coupled Active Inference:

The speaker-listener loop:

1. Speaker has a belief state they wish to communicate
2. Speaker selects words (actions) by evaluating which utterances would most effectively reduce the listener's uncertainty about the intended meaning
3. Listener receives the utterance (observation) and updates their generative model
4. Listener may respond, creating a reciprocal loop

Key philosophical implications:

- **Meaning is not in the words** — meaning emerges from the interaction between the speaker's generative model, the listener's generative model, and the utterance that bridges them
- **Understanding is approximate** — the listener's updated beliefs will never exactly match the speaker's intended meaning; communication aims to minimize the divergence
- **Context is everything** — shared priors (background knowledge, cultural assumptions, conversational history) dramatically affect how utterances are interpreted

Connection to Wittgenstein: This resonates with Wittgenstein's later philosophy — meaning is use, language is a form of life, private languages are impossible. Active Inference formalizes this: meaning arises from shared generative models deployed in practical contexts. A word has meaning because speaker and listener share priors about how that word relates to states of the world.

Connection to Grice: Grice's theory of conversational implicature (the cooperative principle, maxims of quality, quantity, relation, manner) can be derived from EFE minimization. Communicating efficiently means selecting utterances that minimize the listener's uncertainty while respecting shared expectations about conversational structure.

3. Intersubjectivity and Shared Generative Models

Intersubjectivity — the philosophical concept of shared understanding between subjects — is formalized through:

Primary intersubjectivity (Trevarthen): Direct, pre-linguistic attunement between agents — mutual gaze, turn-taking, emotional resonance. Active Inference: basic sensorimotor coupling between dynamical systems, producing generalized synchrony without explicit mentalizing. This begins in infant-caregiver interactions and establishes the foundation for all later social cognition.

Secondary intersubjectivity: Shared attention to objects and events — joint attention, pointing, referencing. Active Inference: coordinated inference about shared environmental states, mediated by communicative actions. The "triangle" of agent-agent-world establishes the conditions for language acquisition.

Tertiary intersubjectivity: Shared concepts, institutions, and cultural knowledge. Active Inference: deeply shared generative models that constitute social reality — money, marriage, law, science. These are collective priors that structure individual inference.

4. Social Ontology and Collective Intentionality

Active Inference has implications for **social ontology** — the study of what social entities are and how they exist:

Searle's institutional facts: Money is money because we collectively believe it is. Active Inference: money is a shared prior in a collective generative model — it reduces prediction error in economic transactions because everyone models it similarly.

Tuomela's collective acceptance: Social institutions exist through collective acceptance. Active Inference: collective acceptance = convergence of individual generative models toward shared priors through iterated communication and mutual updating.

Group agency: Can groups be agents in the Active Inference sense? If a group maintains a Markov Blanket, minimizes free energy collectively, and has an identifiable generative model (organizational strategy, shared beliefs), then it meets the formal criteria for agenthood. This is the "cultural Markov Blanket" thesis — organizations, institutions, and cultures are agents at a higher scale.

5. The Hermeneutic Dimension

Hermeneutics — the philosophy of interpretation — connects deeply to Active Inference:

- **Gadamer's fusion of horizons**: Understanding another person requires a "fusion" of my interpretive horizon with theirs. Active Inference: successful communication requires sufficient overlap in generative models (shared priors) for messages to be interpretable. The "fusion" is the convergence of posteriors through iterated message exchange.
- **The hermeneutic circle**: We understand parts through the whole and the whole through parts. Active Inference: hierarchical generative models implement this circle — top-

down predictions from the whole constrain the interpretation of parts, while bottom-up evidence from parts updates the model of the whole.

- **Prejudice and pre-understanding**: Gadamer argued that all interpretation begins from pre-existing understanding ("prejudice" in a neutral sense). Active Inference: all inference begins from priors. The prior is the formal equivalent of Gadamerian prejudice — it's not bias to be eliminated, but the necessary starting point for interpretation.

Summary

Active Inference provides a formal framework for understanding communication as coupled inference, language as mutual model alignment, and intersubjectivity as the progressive sharing of generative models. The problem of other minds is addressed through higher-order modeling. Social institutions emerge from convergent priors. Hermeneutic interpretation is formalized as hierarchical Bayesian inference from shared priors.

Further Reading

- Vasil, J. et al. (2020). A world unto itself: Human communication as active inference. *Frontiers in Psychology*, 11, 417.
- Friston, K. J. & Frith, C. D. (2015). A duet for one. *Consciousness and Cognition*, 36, 390-405.
- Gallagher, S. & Allen, M. (2018). Active inference, enactivism and the hermeneutics of social cognition. *Synthese*, 195(6), 2627-2648.
- Tomasello, M. (2019). *Becoming Human*. Harvard University Press.
- Veissière, S. P. L. et al. (2019). Thinking through other minds: A variational approach to cognition and culture. *Behavioral and Brain Sciences*, 43, e90.